

On the Limits of Low-Frequency OHLCV Signals in Machine Learning-Driven Portfolio Optimization

A Cost-Aware, Leak-Free Benchmark on Large-Cap US Equities

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Abstract

The application of machine learning (ML) to quantitative finance has grown rapidly, yet financial return series have an extremely low signal-to-noise ratio and are non-stationary, exposing predictive pipelines to severe backtest overfitting. We conduct a rigorous, leak-free, cost-aware stress-test of whether ML models trained only on monthly features derived from daily OHLCV price and volume data can produce economically meaningful, net-of-cost portfolio improvements over disciplined baselines, on a universe of 30 large-cap US equities (2010–2025). A wide spectrum of models, namely regularized linear regressors, tree ensembles, and deep networks (MLP, 1D-CNN, LSTM, CNN-LSTM, Transformer), is evaluated under an anchored walk-forward scheme that retrains at each step, with a pre-specified promotion gate (mean out-of-sample Spearman Rank IC ≥ 0.05 on a 2016 hold-out) that admits only the MLP among the deep models. Promoted forecasts enter a Black–Litterman optimizer as Bayesian views, with Marchenko–Pastur covariance cleaning, dollar-neutral constraints, and realistic transaction costs (10 bps). We report a negative result: in the out-of-sample period (2017–2025) the best model (MLP, Sharpe 0.96) only marginally edges a no-ML Black–Litterman “Prior-Only” baseline (Sharpe 0.94) that shares the same covariance cleaning, while incurring more than three times the cumulative transaction cost (15.13% vs. 4.67%). More strikingly, a naive equally weighted 1/30 portfolio attains a Sharpe of 0.93, essentially tying the entire optimized apparatus. Fama–French five-factor alphas of the active linear and tree strategies are statistically insignificant. The risk-adjusted performance of the pipeline is driven by covariance regularization and Bayesian shrinkage, expressed mainly as superior drawdown control, rather than by the predictive content of the models. We release a fully reproducible pipeline as a baseline for future studies.

1 Introduction

Under the Efficient Market Hypothesis [1], persistent excess returns from public price information should be difficult to obtain in liquid markets. The modern financial machine-learning literature [2] has shown that flexible models can extract non-linear, cross-sectional structure from large predictor sets, reviving the hope of systematic alpha. Yet a large fraction of reported results is contaminated by data-snooping, look-ahead leakage, and the omission of trading frictions [3]. Reported Sharpe ratios that ignore these issues routinely collapse out of sample, and the field suffers from a publication bias toward positive findings. Careful negative results, which delimit where a predictive edge does *not* exist, are therefore valuable: they spare other researchers from chasing exhausted signals.

This paper asks a deliberately narrow and answerable question:

Can machine-learning models trained exclusively on monthly features computed from daily OHLCV price and volume data extract sufficient predictive alpha to generate statistically significant and economically meaningful net-of-cost portfolio improvements in a highly efficient large-cap equity universe, under institutional constraints and trading frictions?

Our hypothesis is that they cannot, and that the risk-adjusted benefits of a realistic pipeline arise instead from robust covariance estimation and Bayesian shrinkage. We test this with an end-to-end, leak-free, cost-aware benchmark and a strict, pre-registered model-promotion gate that prevents non-predictive models from ever reaching the backtest. The design choices are made to be conservative: a highly efficient large-cap universe (a hard test for any technical signal), realistic transaction costs and slippage, anchored walk-forward retraining, and a held-out validation gate.

Contributions.

1. A leak-free, cost-aware benchmark showing that ML models trained solely on monthly OHLCV-derived features do not deliver economically meaningful, net-of-cost improvements over a no-view Black–Litterman baseline on large-cap US equities.
2. A clean performance attribution: by comparing against a Prior-Only baseline that shares the covariance cleaning and optimizer but uses no ML views, and against a naive 1/30 portfolio, we isolate the source of risk-adjusted performance and show it stems from regularization and risk control, not prediction. This reinforces, in a modern ML setting, the classical results of Chopra and Ziemba [4] and DeMiguel et al. [5].
3. A reusable methodological building block: a data-driven calibration of the Black–Litterman view-uncertainty matrix tied to the rolling Information Coefficient, coupled with a disciplined pre-backtest promotion gate.
4. A reproducible open codebase and a frozen data snapshot, providing a baseline against which future studies using higher-frequency or alternative data can be measured directly.

2 Related Work

Classical asset pricing and portfolio theory. Mean-variance optimization [6] is the foundation of quantitative allocation but is notoriously sensitive to estimation error in expected returns and covariances. Chopra and Ziemba [4] show that errors in the mean vector dominate portfolio loss by an order of magnitude relative to errors in the covariance, and DeMiguel et al. [5] find that a naive $1/N$ portfolio is hard to beat out of sample once estimation error is accounted for. These results motivate shrinkage and regularization rather than unconstrained optimization, and they frame our own finding that a 1/30 portfolio is highly competitive.

Bayesian allocation. The Black–Litterman model [7, 8] addresses mean-variance instability by blending a market-implied equilibrium prior with subjective views in a Bayesian update, shrinking extreme allocations toward the market portfolio. A persistent practical difficulty is the specification of the view-uncertainty matrix Ω , which controls how strongly the views move the posterior away from the prior. We address this with a data-driven calibration tied to the rolling skill of each model.

Random matrix theory. Empirical correlation matrices estimated from finite financial samples are dominated by noise. Laloux et al. [9] apply the Marchenko–Pastur law [10] to separate informative eigenvalues from the noise bulk, yielding more stable covariance estimates for optimization. We use this spectral cleaning as the covariance backbone shared by every optimized strategy, which lets us attribute performance differences cleanly to the views alone.

Financial machine learning and leakage. Gu et al. [2] provide a large-scale demonstration of ML for empirical asset pricing, while López de Prado [3] catalogues the methodological pitfalls (leakage, overfitting, and improper cross-validation) that inflate backtested performance. Our work sits in this tradition but emphasizes a careful, negative, replicable benchmark with realistic execution assumptions rather than a new positive claim.

3 Data and Universe

We study 30 large-cap, highly liquid US equities drawn from the S&P 500 and spanning GICS sectors, over 2010–2025. Daily OHLCV data and historical market capitalizations are ingested via Yahoo Finance; the S&P 500 index ($\sim$ GSPC) serves as the market benchmark, and the Fama–French five-factor library [11] supplies the factor returns and the risk-free rate. Data are aligned to the NYSE trading calendar and resampled to monthly frequency. The sample is split into an experimentation period (2010–2016, with 2016 held out for validation) and an out-of-sample backtest (2017–2025).

Scope and survivorship. The universe is a fixed list of companies that are large-cap today, which introduces survivorship bias. We treat this deliberately as a fixed-universe stress-test of technical signals rather than a survivorship-free alpha-discovery study. The bias inflates all strategies comparably, so the relative conclusion, namely that ML views add little over the Prior-Only baseline net of costs, is robust. Absolute outperformance versus the index should be read as caveated, not as a tradeable claim. Restricting the universe to 30 liquid blue chips also keeps a flat 10 bps cost model realistic, since these names absorb institutional-size trades with minimal market impact.

4 Methodology

4.1 Features and target

For each asset and month we compute a panel of technical features from daily OHLCV bars. These span six families: (i) returns and momentum across multiple horizons; (ii) intraday price-action metrics derived from the open-high-low-close geometry; (iii) range-based volatility estimators such as the Parkinson [12] and Rogers–Satchell estimators; (iv) volume and liquidity measures, including the Amihud illiquidity ratio [13]; (v) efficiency, persistence, and higher-moment statistics (skewness and kurtosis); and (vi) CAPM-derived systematic and idiosyncratic components (rolling beta and residual alpha). The prediction target is the next-month log-return,

$$Y_t = \ln\left(\frac{P_{t+1}}{P_t}\right). \quad (1)$$

4.2 Preprocessing and leakage control

All cross-sectional preprocessing is fit on past data only. Features are winsorized at cross-sectional quantiles to limit the influence of outliers, then standardized using rolling statistics, so that no

future information enters the scaling of any training sample. Highly collinear features are pruned with variance-inflation-factor diagnostics to stabilize the linear models and the covariance estimates. This discipline is the practical answer to the leakage critique of López de Prado [3]: the entire pipeline is causal with respect to the forecast boundary.

4.3 Models

We evaluate three model families. The linear family comprises ordinary least squares and its regularized variants (Lasso, Ridge, ElasticNet), which are interpretable and naturally resistant to overfitting in low signal-to-noise regimes. The tree family comprises Random Forest (bagging) and XGBoost (gradient boosting [14]), which capture non-linear interactions. The deep family comprises a multilayer perceptron (MLP), a 1D temporal CNN, an LSTM, a CNN-LSTM hybrid, and a Transformer [15]. Networks are trained with the Huber loss (robust to return outliers) and the AdamW optimizer under a cosine-annealing learning-rate schedule.

4.4 Walk-forward validation, hybrid retraining, and the promotion gate

We use an anchored walk-forward scheme that retrains at each monthly step on an expanding window. To control hyperparameter-search overfitting and computational cost, a high-dimensional Optuna [16] search (tree-structured Parzen estimator with successive-halving pruning) is run once per year, in December, and the chosen hyperparameters are then re-fit on the expanding window during the intermediate months. This hybrid reduces compute by roughly 90% relative to monthly re-tuning while preserving out-of-sample integrity.

Before any model may enter the backtest, it must clear a pre-specified *promotion gate*: a mean out-of-sample Rank IC of at least 0.05 on the 2016 hold-out window. The Information Coefficient at month t is the Spearman rank correlation between predicted returns $\hat{R}_{i,t+1}$ and realized returns $R_{i,t+1}$ across the N stocks, computed as the Pearson correlation of their tie-corrected ranks,

$$\text{Rank IC}_t = \frac{\text{cov}(\text{rank}(\hat{R}_{\cdot,t+1}), \text{rank}(R_{\cdot,t+1}))}{\sigma_{\text{rank}(\hat{R})} \sigma_{\text{rank}(R)}}, \quad \overline{\text{IC}} = \frac{1}{T} \sum_{t=1}^T \text{Rank IC}_t. \quad (2)$$

In the implementation, Eq. (2) is evaluated with `scipy.stats.spearmanr` for the shallow models and with an equivalent differentiable Pearson-on-ranks routine (using average-tie ranks) for the neural networks, so that the same statistic drives both validation and the gate. The threshold of 0.05 reflects the fact that a cross-sectional IC in this range is a weak but genuine signal in equity markets; models below it are presumed to carry no exploitable information.

4.5 Allocation: Black–Litterman with covariance cleaning

Promoted forecasts enter the Black–Litterman model [7, 8] as views Q . The posterior mean return vector is

$$\hat{\Pi} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau\Sigma)^{-1} \Pi + P^\top \Omega^{-1} Q], \quad (3)$$

where Π is the market-implied equilibrium prior (from reverse optimization on market-cap weights), P the link matrix, and τ a prior-scaling parameter. The diagonal view-uncertainty matrix Ω is calibrated dynamically from each model’s rolling Rank IC: a higher recent IC, with a higher signal-to-noise ratio across the IC history, produces lower view uncertainty and therefore greater posterior weight on the model’s views. This ties the confidence placed in a model directly to

its demonstrated out-of-sample skill. The covariance Σ is cleaned with random matrix theory: eigenvalues falling within the Marchenko–Pastur band [9, 10],

$$\lambda_{\pm} = \sigma^2 \left(1 \pm \sqrt{N/T}\right)^2, \quad (4)$$

are treated as noise and replaced by their average, preserving total variance while removing spurious correlation structure.

4.6 Backtesting engine and baselines

Portfolios are solved monthly under institutional constraints (dollar neutrality, a gross-leverage cap of 2.0, and per-position bounds) using the ECOS conic solver [17] in CVXPY. Transaction costs are modeled at 10 bps (5 bps commission plus 5 bps slippage, in the spirit of Almgren and Chriss [18]), applied to traded turnover, with weight drift between rebalances and a self-financing structure that yields net excess returns. We benchmark every active strategy against five baselines: the S&P 500 index; an equally weighted 1/30 portfolio; unregularized mean-variance optimization (MVO); the Prior-Only Black–Litterman portfolio (same covariance cleaning and optimizer, no ML views); and a random-view control that injects pure noise as views.

4.7 Statistical evaluation

Because overlapping monthly returns are autocorrelated and heteroskedastic, ordinary standard errors are biased; we therefore report Newey–West HAC [19] t -statistics. Risk-adjusted alpha is assessed with Fama–French five-factor regressions [11], and path robustness with Politis–Romano stationary-bootstrap [20] Monte-Carlo simulation (10,000 resampled paths per strategy), from which we read terminal-wealth, Sharpe, and drawdown distributions.

5 Results

5.1 Promotion gate

Among the deep models, only the MLP cleared the gate (mean Rank IC 0.061); the CNN (0.038), LSTM (0.029), CNN-LSTM (0.040), and Transformer (0.029) failed, consistent with overfitting in a data-scarce panel and the absence of peak-validation checkpointing. The promoted set therefore comprises the regularized linear models, the two tree ensembles, and the MLP.

5.2 Headline performance

Table 1 summarizes the 2017–2025 backtest, and Figure 1 shows the out-of-sample equity curves of the MLP and Prior-Only strategies. Three regularities stand out.

First, the optimized long-short strategies produce visually near-identical equity curves, because every allocation is dominated by the same Marchenko–Pastur-cleaned covariance and the same market-cap equilibrium prior; the ML views perturb, but do not reshape, these weights. The MLP and Prior-Only curves in Figure 1 are almost indistinguishable.

Second, on a risk-adjusted basis the Prior-Only baseline (Sharpe 0.94) and the MLP (Sharpe 0.96) lead, ahead of the passive S&P 500 (0.72), while the individual linear and tree models cluster well below (Sharpe 0.50 to 0.58), dragged down by the turnover their active views generate, and the naive MVO portfolio collapses (Sharpe -0.29 , a total return of -22.47%). The active machine-learning

Table 1: Out-of-sample results (2017–2025), net of 10 bps transaction costs. Best value per column in **bold**. ElasticNet, XGBoost, and Random Forest are shown as the representative linear and tree models; the remaining linear models fall in the same 0.50–0.58 Sharpe cluster.

Strategy	Cum. Return	CAGR	Sharpe	Sortino	Max Drawdown
S&P 500 Index	200.39%	13.00%	0.72	1.10	−33.92%
Equally Weighted (1/30)	323.41%	17.39%	0.93	1.50	−34.46%
Mean-Variance (MVO)	−22.47%	−2.78%	−0.29	−0.38	−40.59%
BL Prior-Only (No ML)	236.94%	14.45%	0.94	1.66	− 25.80%
BL MLP (Active)	244.89%	14.75%	0.96	1.70	−25.82%
BL ElasticNet (Active)	129.89%	9.70%	0.58	0.97	−25.16%
BL XGBoost (Active)	126.31%	9.51%	0.55	0.93	−27.06%
BL Random Forest (Active)	109.21%	8.55%	0.50	0.81	−27.46%

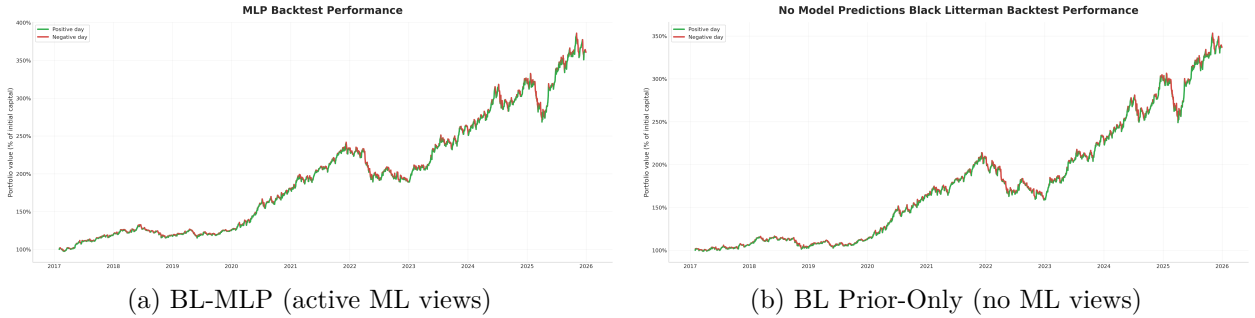


Figure 1: Out-of-sample cumulative performance (2017–2025), as a percentage of initial capital. The active-ML (a) and no-ML (b) portfolios trace nearly identical paths, illustrating that the risk-management layer, not the ML views, drives the result. See Table 1 for the full comparison.

models never convincingly beat the no-model Prior-Only baseline net of costs, and only the MLP edges it at all, by a slim and cost-eroded margin.

Third, and most tellingly, the equally weighted 1/30 portfolio achieves a Sharpe of 0.93 and the highest raw return in the study (323.41%), essentially tying the entire optimized apparatus on a risk-adjusted basis. This is a direct echo of DeMiguel et al. [5]: naive diversification is extremely hard to beat. The optimized Black–Litterman strategies do retain one genuine advantage, but it is in *risk control*, not return: their maximum drawdowns (−25.8%) are markedly shallower than the fully market-exposed 1/30 and index portfolios (−34 to −36%), and they are dollar-neutral with near-zero market beta rather than carrying a beta of 1.0.

5.3 The decisive role of transaction costs

Costs explain why the active models underperform the Prior-Only baseline. The Prior-Only portfolio trades slowly and accumulates only 4.67% in cumulative transaction costs over the period, whereas the MLP, whose active views drive much higher turnover, accumulates 15.13%, and XGBoost is the most expensive strategy in the study at 17.46%. The 1/30 portfolio trades cheaply by construction. The active models’ gross predictive edge, already small, is therefore almost entirely eroded by trading friction, leaving net Sharpe ratios in the 0.50 to 0.58 range for the individual linear and tree models.

5.4 Statistical significance and factor attribution

HAC-adjusted t -statistics of net return drifts are significant for Prior-Only (3.26), MLP (3.19), and Equally-Weighted (4.18), but fall to the 1.8 to 2.1 range for the individual linear and tree models, so their net returns are only weakly distinguishable from zero. The Fama–French five-factor regressions tell the same story: the annualized alphas of the linear and tree strategies (2 to 3%) carry p -values around 0.5 to 0.7 and are statistically insignificant, meaning none delivers risk-adjusted return beyond passive factor exposure. The active models load on a small positive market beta (about 0.32 to 0.35) with negative value and investment tilts, a near-uniform factor fingerprint across the whole active cohort.

5.5 Where the performance comes from

Because the Prior-Only baseline shares the covariance cleaning and the optimizer but uses no ML views, its near-identical risk-adjusted performance localizes the source of utility to the risk-management layer (Marchenko–Pastur cleaning and Black–Litterman shrinkage) rather than to model prediction. Two further controls sharpen this. The MLP’s cross-sectional Model IC decays from about 0.06 in validation to about 0.02 live, confirming that the residual edge is small and unstable. And the random-view control, which feeds pure noise into the same optimizer, still produces positive returns: it establishes a noise floor showing that the apparatus makes money from risk management even when the views contain no information whatsoever.

5.6 Monte-Carlo robustness

The stationary bootstrap corroborates the point-estimate reading. The Sharpe-ratio distributions of the active models overlap heavily with that of the Prior-Only baseline, and the wide dispersion of terminal-wealth cones overwhelms the small differences in their centres. No active model converts its nominal point-estimate lead into a reliably repeatable edge across resampled paths.

6 Discussion: Practical Implications

Beyond the headline finding, the study yields two practitioner-facing lessons. First, on methodology: the gap between the gross and net performance of the active models is a concrete demonstration that backtests omitting realistic turnover and transaction costs are misleading, and that a pre-specified promotion gate is an effective, cheap guard against advancing non-predictive models into an expensive backtest. The same architecture that reports a high Sharpe on paper can be revealed as cost-dominated once friction and a held-out skill check are imposed. For allocators, the operational implication is that robust covariance estimation and disciplined shrinkage deliver most of the realisable utility, and are far more reliable than a marginal, unstable forecasting edge.

Second, on computational cost: the hybrid retraining scheme, which restricts the expensive Optuna hyperparameter search to an annual cadence while re-fitting parameters monthly on the expanding window, reduces compute (and the associated energy use) by roughly 90% relative to retuning at every rebalance, with no loss of out-of-sample integrity. In an era of growing scrutiny of the energy footprint of machine learning, this is a practical template for keeping rolling walk-forward research tractable and efficient without compromising methodological rigour.

7 Limitations

The universe is small (30 names), US large-cap, and monthly: the most informationally efficient regime, and not generalizable to small-caps or higher frequencies. The fixed universe carries survivorship bias, bounded in Section 3. The deep-model outcome is partly capacity-bound and implementation-bound: the panel (about 2,160 observations) is data-scarce for sequential networks, and training lacked peak-validation early-stopping checkpointing, so the result should be read as “not supported in this setup” rather than as a general impossibility. The promotion gate uses a single hold-out year.

8 Future Work

The natural extensions follow directly from these limitations. Higher-frequency limit-order-book data would capture microstructure dynamics (order-flow imbalance, spread variation) that monthly bars smooth away, and would invite continuous-time tools such as Hawkes processes and stochastic differential equations. Broadening the predictor space to macroeconomic series and alternative data (news and filings sentiment) could resolve cross-sectional dispersion that price-only features miss. Expanding the universe to the full S&P 500 or Russell 3000 over several decades would supply the sample sizes that deep sequential models require, paired with proper EMA-IC-based early stopping. Finally, moving from point forecasts to predictive distributions (quantile or mixture-density models) would allow tail-risk-aware optimization and an end-to-end Monte-Carlo stress test that re-generates signals and re-runs the optimizer on each path.

9 Conclusion

On large-cap US equities at monthly frequency, features computed from daily OHLCV data carry a signal too weak and unstable to generate robust, cost-justified alpha. A disciplined pipeline still produces attractive risk-adjusted performance, but that performance is driven by covariance regularization and Bayesian shrinkage, expressed as superior drawdown control, not by ML prediction; indeed a naive 1/30 portfolio ties the optimized strategies on Sharpe. Documenting this negative result carefully, with realistic costs and a pre-registered gate, is the contribution: it marks where a predictive edge is not found, and provides a reproducible baseline for studies using higher-frequency, alternative, or macro data.

Reproducibility

The complete pipeline, a frozen data snapshot, and the full thesis are archived with a permanent DOI, and the code is released under the MIT license. The global random seed is fixed; because Yahoo Finance adjusted prices are recomputed on later corporate actions, exact reproduction requires the frozen snapshot rather than a fresh download. See the repository `README` and `DATASET.md` for details.

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